

WELD DEFECT INSPECTION REPORT

AI-Powered Visual Inspection System

Report ID	RPT-2026-00001
Sample ID	qwe
Scan Number	#1
Scan Date & Time	02 May 2026, 12:41
Operator	ghhfd
Location / Site	Welding Bay 3, Plant A — Pune
Specimen Type	Pipe / Cylindrical
Joint Configuration	Single V Butt Joint (Plate)
Material	Stainless Steel
Specimen Size	250 x 400 mm
Welding Process	SMAW (Shielded Metal Arc Welding)
Industry / Application	Oil & Gas
Detection Device	Inspection Camera

1. SCAN SUMMARY

The AI-powered visual inspection of sample **qwe** (Scan #1) was performed on **02 May 2026, 12:41** using the AI model (version v1 (Simulation)). The scan covered the **CAP** side of the weld on a **Pipe / Cylindrical** specimen.

1	1	MEDIUM	63.2%
Total Defects	Defect Types	Highest Severity	

Defects Detected:

#	Defect Type	Count	RPN	Severity	Surface Visible	Accuraccy
1	Porosity	1	210	MEDIUM	Yes	63.2%

2. DETAILED DEFECT FINDINGS

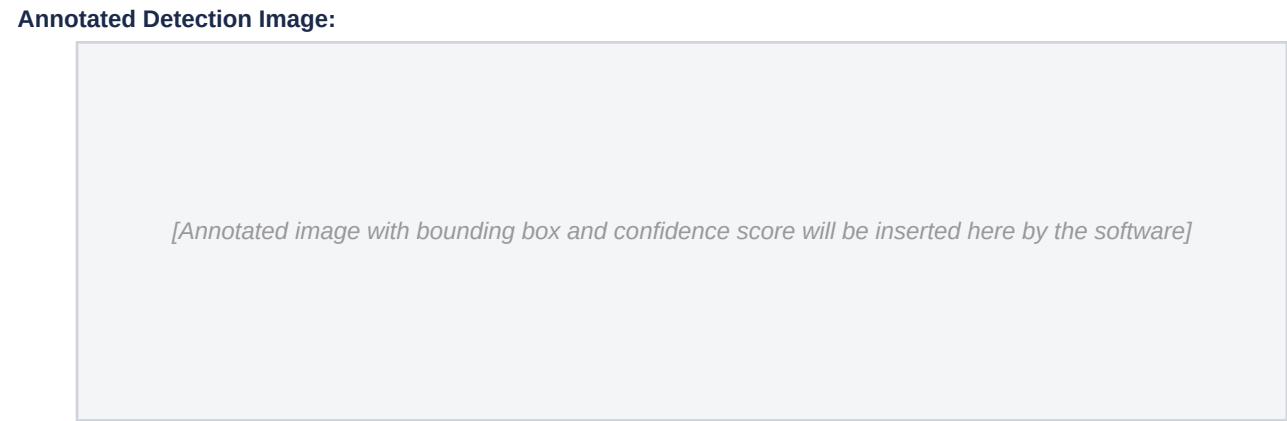
2.1 Porosity	MEDIUM	RPN: 210
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Description:
Gas pockets trapped within or on the surface of the weld metal. Appears as spherical or elongated voids caused by gas entrapment during solidification.

Detection Details:

Instances Detected:	1
Accuracy:	63.2%
Surface Visible:	Yes

Primary Causes:
Gas entrapment, contamination (oil, moisture, rust), improper shielding gas flow, excessive travel speed, wet electrodes.



Applicable Standards:
ISO 5817 Level B: max 3% area; API 1104: max individual pore diameter limits apply; AWS D1.1: size and distribution limits.

Recommended Action:
Evaluate pore size and distribution. If within acceptance criteria, document and monitor. If exceeding limits, grind and re-weld affected area. Verify shielding gas setup.